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A Novel Efficient Approach for Screen Image Classification in Remote Display Protocol

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***Abstract*—In remote display protocols, screen image compression plays an important role to improve quality of experience (QoE) of users and reduce the bandwidth consumption. Not all image elements on the display have the same type, so it is wise to apply a screen image classification for compression decision. In this paper, we propose a novel efficient approach for screen image classification that separate the captured screen into 2 types of blocks: text and non-text block. Our method is a 2-stage process that is different from other works because of the appearance of text localization in the screen image as the first stage. This text localization is mainly based on edge feature and morphological operation in which we experiment with many kinds of edge detection methods. Then block-based classification categorizes the screen image blocks based on the positions of the detected text regions. The experimental results of high accuracy rate and low time consumption state our method is efficient in remote display protocol.**

***Keywords*—remote display protocol; screen image classification; edge detection; text localization.**

I. Introduction

In remote display protocols, screen compression technologies are made used to reduce the often limited network bandwidth as well as quality of display to satisfy demands of user. The server may compress a screen image with some coding before sending the image across the network, and the client then performs the corresponding decompression operation after receiving the image. Based on the observation that not all image elements on the display have the same type, it makes no sense to use the same coding for everything - that would use excessive network bandwidth and give lesser user experience. So, it is wise to apply a screen image classification to determine preferred type of compressor for the image content. In particular, the server classifies the screen compound image, which are often mixed with textual, graphical or pictorial contents, and based on that classification determines a coding with which to encode each image content, and a fidelity to use on the encoding. The server performs that encoding with the coding and the fidelity, and then sends this encoded image to the client. So, we realize that screen image classification is a key challenge in compression decision, not only choosing suitable codings for screen image contents, but also the quality setting to use for a particular coding. PCoIP [1], which is invented by Teradici Corp, is a multi-codec protocol that continuously analyzes and decomposes the image elements - graphic, text, icons, video, etc. - and compresses them with the right coding for each pixel. This enables efficient transmission and decoding and saves bandwidth while delivering best user-experience.

In this paper, we propose a novel efficient approach for screen image classification that separate the screen compound image into text and non-text blocks. After doing the classification, we can apply the most suitable compression format for each kind of block, because text block needs more qualitative vision and is often updated slowly.

The remainder of the paper is organized as follows. Section 2 reviews some related work. Section 3 gives a detail description about our proposed method. Then, some experimental results is represented in section 4, followed by conclusion in section 5.

II. Related work

Several works have been contributed to screen image classification. Keslassy et al. [2] proposed transform coefficient likelihood (TCL) schemes in order to separate the textual and graphical portions of a compound screen image by computing DCT coefficient of an 8 x 8 block. Wenpeng Ding et al. [3] classified the screen image into four different types - smooth blocks, text blocks, hybrid blocks and picture blocks with a fast and effective block-based classification algorithm based on two features: histogram and gradient of the block. Said et al. [4] proposed a color-based classification by counting the number of different colors in a block. While Yanfei Shen et al. [5] classified the screen image into 3 types, text/graphic, pictorial and hybrid blocks based on the characteristics of histogram distribution and the number of colors. These kinds of algorithms belong to block-based approach, where the compound image is divided in blocks of a certain size and

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a classification is applied to the blocks to categorize them into different types. These classifications are often based on some simple characters of images such as color number, histogram, gradient and so on.

In this paper, we propose a novel approach for classifying the screen image into 2 types of blocks: text and non-text block. Based on the observation that text in an image is not a single character but many characters, which are arranged next to each other horizontally to form text lines and text lines are arranged from top to bottom with a small gap between them to form paragraphs, so text often exists in a screen as text region that spread through many small blocks. Therefore, we propose a multi-stage screen image classification. We perform text localization in the screen image to detect text regions as the first step of our method. Then block-based algorithm will be applied to separate image into text and non-text blocks. The appearance of text localization is what makes our method different from other works.

III. Methodology

As we mentioned above, our proposed method for screen image classification is a 2-stage process. The diagram of our method is illustrated in Figure 1.

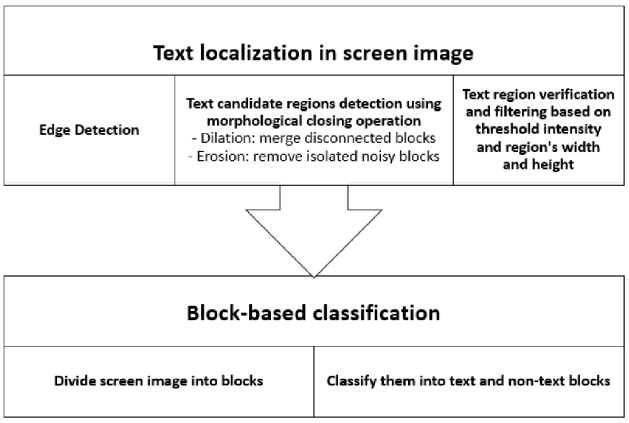

Fig. 1. Screen image classification diagram

The first stage is text localization in the screen image which detects the regions that contain text. It consists of 3 main steps. Based on the observation that edge is a good feature to recognize text, first, an edge detection method is applied to input grayscale image and then thresholding to generate a binary edge image. There are many kinds of edge detection method, so we make use some of them and make a comparison about their efficiency in text detection. In the second step, a morphological closing operation [6] will be performed on the resulting binary image from the previous step to form text candidate regions. Morphological closing operations include two fundamental operations from the field of mathematical morphology, dilation and erosion. With the help of dilation operation, we can group disconnected blocks into potential text regions and then isolated noisy blocks can be discarded by performing erosion operation. Finally, we verify the candidate regions by computing the intensity of each region and compare it with a predefined threshold for text. We can also eliminate very small regions by specifying minimum accepted height and width of text region.

After we detects the text regions in the screen image, the output image will be processed by the second stage, which is block-based classification. In this stage, we divide the input image into blocks and classify them into text and non-text blocks. This can be done by comparing the position of the text regions and all the image blocks. If a block that contains a part of some text region, it is determined text block, otherwise it is non-text block.

A. Text localization in the screen image

1) Edge Detection: Edges are significant local changes of intensity which characterize the boundary between two different regions in an image. Text regions typically have a large number of discontinuities, e.g. transitions between text and background since text portion commonly has high intensity contrast with respect to its background. So, edge feature can be considered as a good parameter to divide image into picture and text blocks. There are many ways to perform edge detection. The most common method for edge detection is to calculate the differentiation of an image. It can be grouped into two main categories: Gradient-based edge detection (first-order derivatives) and Laplacian-based edge detection (second-order derivatives). The former method detects edges by computing the gradient of the image intensity at each point, which points in the direction of largest possible intensity increase from light to dark pixel area and the rate of change in that direction. In other word, the gradient shows possibility to become an edge of the image. Currently, we use Sobel operator [7] and Canny optimal edge detection [8], which belong to this method. The latter method detects edges by computing the Laplacian and we use Laplace operator [9].

First and second derivatives are known as linear detectors to locate edge map. Another approach for edge detection is to use morphological operations, which rely only on the relative ordering of pixel values, not on their numerical values and thus it is known as non-linear method. The specific operator used for edge detection is morphological gradient [10], which is the difference between the dilation and the erosion of an image.

2) Text candidate regions detection using morphological closing operation: In this step, we use morphological closing operator, which is simply dilation followed by erosion using the same structuring element for both operations. In order to detect text candidate regions, first, a dilation operator can easily connect closed edges of regions of foreground pixels in the image and therefore help to group them into potential text regions. Based on the observation that characters are typically arranged next to each other horizontally to form text lines, we currently use a rectangle structuring element of size 10x1. This particular size allows to merge disconnected components into horizontal lines to form text candidate regions. After that, we apply erosion operation to filter the small isolated noisy regions in the dilated image. Besides, performing an erosion on the image after the dilation will also reduce some of the effect on the original boundary of the regions caused by dilation. Finally, the output is a binary image that contains text candidate regions.

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3) Text region verification and filtering based on threshold intensity and region's width and height: After finding text candidate regions, we can verify them by calculating intensity, denoted as I in each region and compare it with a predefined threshold T for text. First, we find the external contours of the image corresponding to each region, and define the bounding rectangle of each contour. The intensity I of each candidate region is computed by dividing the number of non-zero pixels (white pixels) in the bounding rectangle R of the region by the area of R that equals to the multiplication of R's height and width. A text region is required to have the intensity I greater than threshold T.

In order to have better result, we can also add some more conditions on the size of region to eliminate very small regions. For example, we constrain the width and height of the text region to be greater than some threshold. If a region satisfies these two conditions of the intensity I and the region's size, it is determined text region, otherwise it belongs to non-text region. The whole process for text localization in screen image is presented in the Algorithm 1.

Algorithm 1: Text localization in screen image

1. Apply an edge detection method
2. Thresholding to obtain binary screen image
3. Apply morphological closing operation to detect text candidate region
4. Set threshold T for text region
5. Set threshold H and W for region's height and width respectively
6. Find external contours of the screen image corresponding to each region
7. For each contour of region
   7.1. Define bounding rectangle R
   7.2. Compute intensity I in each region

$$I = \frac{number_of_white_pixels_in_R}{area(R)} \quad (1)$$

   7.3. Text region verification and filtering
   If($I > T$ AND $R.height > H$ AND $R.width > W$) then region is text region; otherwise region is non-text region

The output of the text localization algorithm is an image with bounding rectangles of the determined text regions, each bounding rectangle is specified by its top-left corner point (x,y) and its height and width as can be seen in Figure 2. In the next stage, we will separate the image into blocks and classify them into text and non-text blocks.

B. Block-based classification

In this stage, we separate the screen image into non-overlapping m*n blocks (image tiles). The screen image is now considered as a matrix of blocks, arranged in rows and columns according to x-dimension and y-dimension respectively. The number of rows and columns can be easily computed by ceiling values when dividing the image's height by block's height (n) and dividing the image's width by block's width (m) respectively. We can treat the elements of this matrix as a one-dimensional array, denoted by $ITArray[]$ whose block index can be accessed by the formula: $rowIndex * number_of_columns + columnIndex$. Initially, the value of all elements in $ITArray[]$ are set to 0, then if a block at index k contains text, we set $ITArray[k] = 1$ and that block is determined text block. This can be done based on the position of the text regions we detect in the previous stage. In particular, for each text region, its bounding rectangle R may spread through many blocks, so we calculate the range of row indexes (top→bottom) and column indexes (left→right) of all blocks in the screen image to which R belongs and set the value of those blocks to 1. After the classification, all the blocks in the screen image that have value 1 are determined text block, the others that have value 0 are determined non-text blocks. The block-based classification is presented in Algorithm 2. The example can be seen in Figure 2.

Algorithm 2: Block-based classification

1. Divide screen image into non-overlapping m*n blocks
2. Set all block values in $ITArray[]$ to 0
3. For each bounding rectangle R of text region
   3.1. Calculate the range of row indexes and column indexes of blocks to which R belongs
   3.2. For i from top to bottom
      For j from left to right
         $ITArray$[i*number_of_columns+j]=1

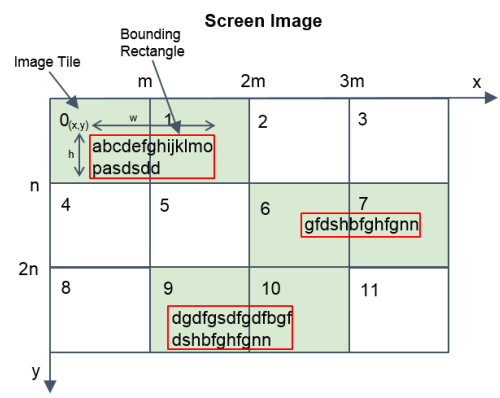

Fig. 2. Block-based classification

IV. Experimental Results

The proposed method is implemented by using Visual Studio 2010 and open source openCV, which is a library of programming functions for real time computer vision. We compare the efficiency of our method when applying various edge detection in the first step of the text localization in the screen image. Currently, we use Sobel, Laplace, Canny and morphological gradient for edge detection. These algorithms are the built-in functions in openCV. Our input image is the website (naver.com) of the size 640x512 which consists of text, image. The screen image is separated into non-overlapping 16*16 blocks. Then, we calculate the accuracy rate by dividing the number of truly detected text/non-text blocks by the number of blocks. The accuracy rate (AR) can be written by

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the following formula:

$$AR = \sum_{i=1}^{number_of_blocks} \frac{accuracy_block(i)}{number_of_blocks} \quad (2)$$

where, $accuracy_block(i) = 1$ when $block(i)$ is text/non-text and truly detected text/non-text block. Otherwise, $accuracy_block(i) = 0$ when misdetection occurs, which means $block(i)$ is text/non-text block but detected non-text/text block. The $number_of_blocks$ can be easily computed based on the image size and block size. Figure 5 shows the results when applying each of 4 edge detections mentioned above in our method.

We continuous to test with other websites such as: Gmarket, Youtube,... and then compute the average of accuracy rate. Figure 3 shows that the average accuracy rate of our method is the best when we apply morphological gradient for edge detection.

Another metric for the efficiency of our method is time consumption, which is one of the important factor that directly affect QoE. This should be as small as possible. Figure 4 shows that the best case is also when we apply morphological gradient for edge detection in our method. By combining these results, we can easily see that morphological gradient is the best edge detection method to apply in our screen image classification.

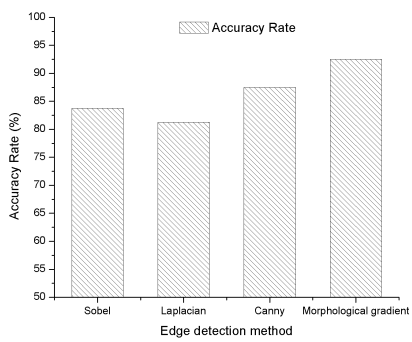

Fig. 3. Accuracy rate of our method when applying various edge detection

Fig. 4. Time consumption of our method when applying various edge detection

Fig. 5. Results of our method

V. Conclusion

In this paper, we propose a novel efficient approach for screen image classification in remote display protocol. Our method is different from other works because of the appearance of text localization to detect text regions exactly in the first stage. Then block-based classification algorithm is simply based on the position of text regions to separate the screen image into 2 types of blocks: text and non-text block. In the text localization process, we make use of edge detection and morphological closing operation and we also experiment with various edge detection algorithms to find the most suitable for our method, which is morphological gradient that give the most accuracy rate and the least time consumption. Based on the experimental result of high accuracy rate, we see our method is purely good screen image classification to apply for remote display protocol with low complexity computation.

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